

Evidence-Gap Assessment: TEC.3 System/ Software Requirements Engineering

Base Practice	Status	Rationale
Prepare for system/ software requirements definition	EvidenceStatus.STRONG_EVIDENCE	The assessed claims demonstrate that the functional boundaries and requirements strategies have been clearly outlined and adhered to, particularly shown in how OptArrow integrates Apache Arrow for cross-language data interchange (Claim C-001) and its compliance with specific standards (Claim C-002). These support a structured preparation of requirements definitions.
Define system/ software requirements	EvidenceStatus.STRONG_EVIDENCE	Claims such as C-005, which involves secure data handling implementation, and C-007, concerning the HiGHS library for optimization, reflect meticulous definition of function, implementation constraints, and associated risk factors, which align with the intent of comprehensive requirements definition.
Analyse system/ software requirements	EvidenceStatus.PARTIAL_EVIDENCE	The claims largely reflect understanding and implementation of system requirements, yet specific evidence related to continuous feedback and review processes in place is limited. The justification challenges (Claim C-003) highlight areas needing refinement in requirements analysis.
Manage system/ software requirements	EvidenceStatus.PARTIAL_EVIDENCE	There is clear evidence of maintaining necessary requirements artifacts, as seen in claims like C-004's task execution support. However, while key artefacts are mentioned, there is less concrete evidence of ongoing traceability and stakeholder agreement on the requirements.